

BERTastic at SemEval-2024 Task 4: SOTA Multilingual Propaganda Detection in Memes via Zero-Shot Learning with VLMs

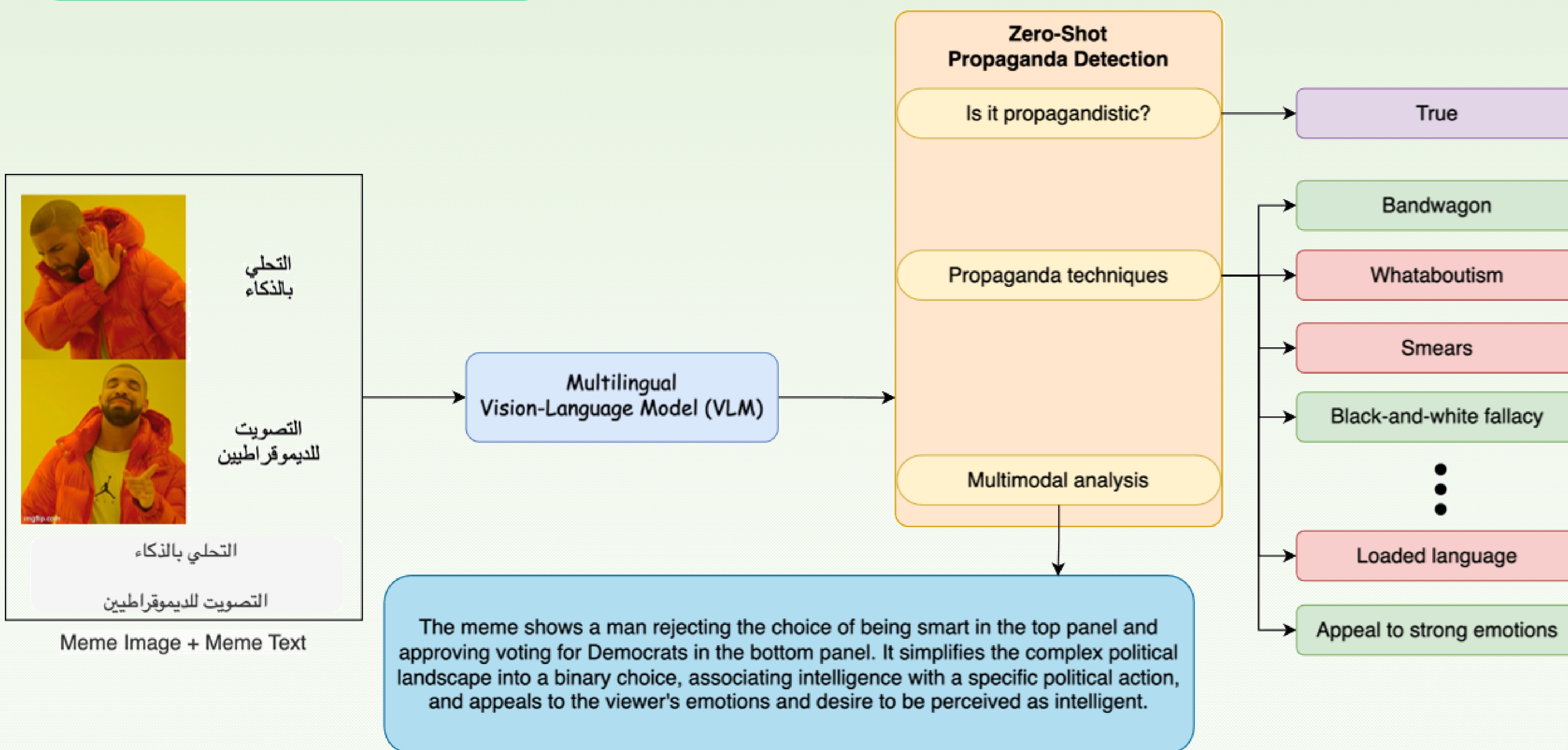
Tarek Mahmoud – tarek.mahmoud@mbzuai.ac.ae
Preslav Nakov – preslav.nakov@mbzuai.ac.ae

Contributions

1. Present a **zero-shot** learning approach achieving **SOTA average performance** on multilingual, multimodal propaganda detection in memes with a macro F1 score of 66.7%
2. Present an **early fusion** approach for identifying persuasion techniques in memes in a multilabel classification setting **outperforming all other approaches in average hierarchical precision** with an average score of 77.66%

Zero-Shot Learning with VLMs

Binary Classification Subtask



Results

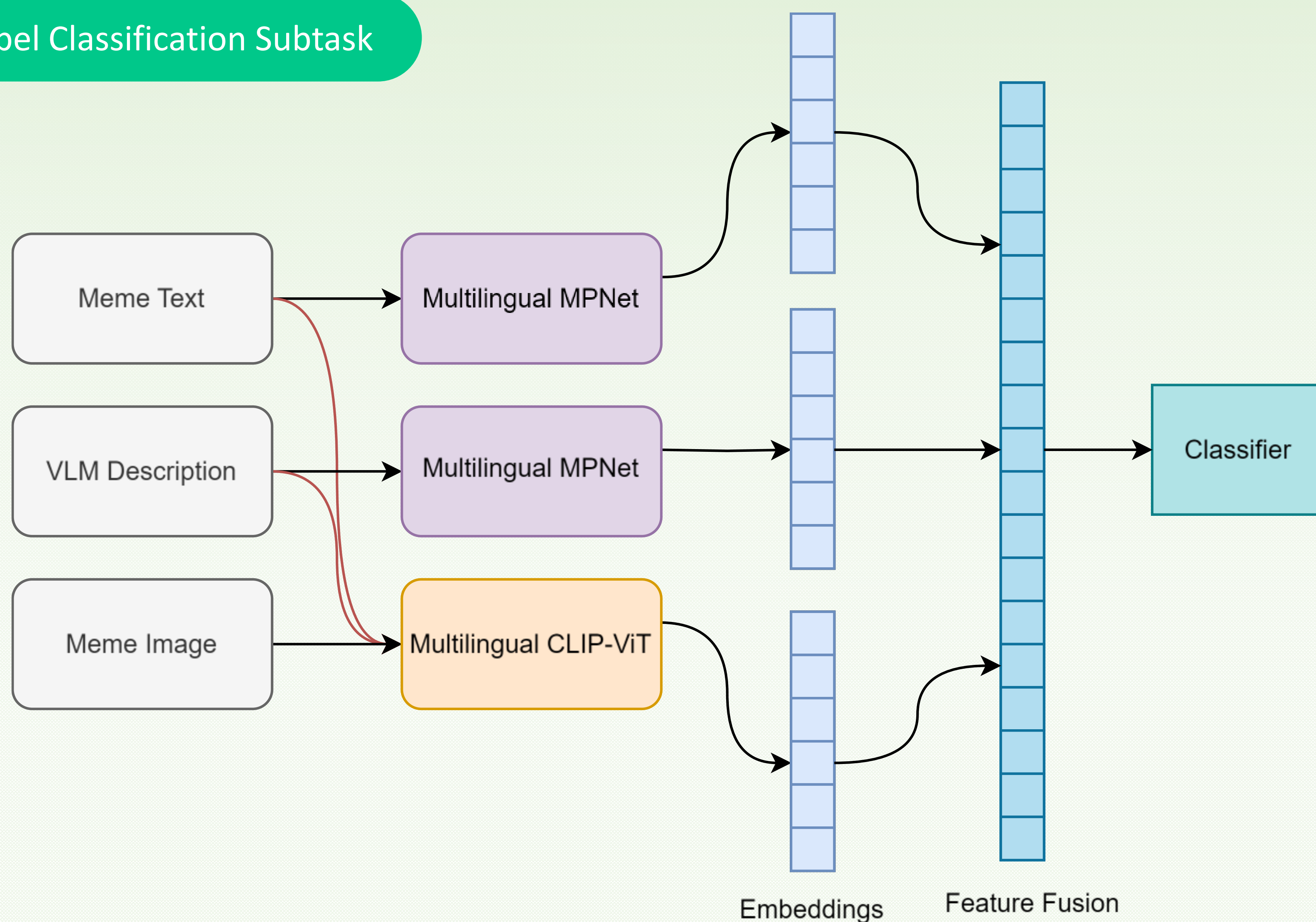
Team	Avg F1	Macro F1			
		Arabic	English	Bulgarian	North Macedonian
BERTastic (ours)	66.67	60.28	71.58	66.21	68.63
BCAmirs	65.66	61.49	80.34	64.72	56.1
NLPNCHU	63.51	58.52	78.8	64.71	52.0
Snarci	62.52	55.54	79.86	66.78	47.92
LMEME	60.85	36.2	81.03	67.1	59.08
SheffieldVeraAI	56.16	61.03	64.2	53.62	45.79
BDA	56.1	50.97	79.29	50.62	43.54
DUTIR938	54.52	46.89	80.91	43.41	46.88
HierarchyEverywhere	52.92	56.2	56.31	48.55	50.62
SuteAlbastre	52.05	50.07	80.96	59.45	17.7
Hidetsune	48.95	52.82	71.35	32.67	38.94
IITK	47.71	46.71	48.34	47.26	48.55
nowhash	46.47	49.83	49.84	43.36	42.86
MemeSifters	45.71	55.65	–	61.14	66.03
UMUTeam	19.66	–	78.66	–	–
TUMnlp	19.6	–	78.41	–	–

Additional Experiments

Method	Training Data	Macro F1
XGBoost	CLIP-ViT embeddings	64.34
LightGBM	CLIP-ViT embeddings	70.1
SVM	CLIP-ViT embeddings	71.91
Zero-shot VLM	Meme image & text	75.08

Early Fusion

Multilabel Classification Subtask



Results

Team	Avg P	Arabic			English		
		F1	P	R	F1	P	R
BERTastic (ours)	77.66	38.82	61.29	28.41	61.34	81.58	49.14
Baseline	76.11	48.65	65.0	38.87	44.71	68.78	33.12
HierarchyEverywhere	70.91	43.69	50.99	38.21	74.59	86.68	65.46
BCAmirs	69.75	52.61	55.31	50.17	70.5	78.37	64.06
NLPNCHU	69.73	48.32	59.47	40.70	70.68	78.16	64.5
SuteAlbastre	58.48	51.61	46.94	57.31	68.48	71.78	65.47
IITK	57.65	45.54	45.73	45.35	63.6	76.29	54.54
BDA	49.15	41.64	38.25	45.68	50.39	51.48	49.34
LomonosovMSU	19.79	–	–	–	65.61	79.15	56.02
TUMnlp	19.52	–	–	–	67.72	78.07	59.79
UMUTeam	19.19	–	–	–	69.0	76.76	62.67
Pauk	18.63	–	–	–	67.53	74.5	61.75
CodeMeme	15.16	–	–	–	66.62	60.66	73.88
WhatsaMeme	7.84	–	–	–	36.59	31.34	43.96

Additional Experiments

Method	Training Data	F1	P	R
MPNet + FFN	VLM descriptions	36.34	57.62	26.55
DeBERTa-V3-Large	Meme text & VLM descriptions	41.34	31.33	60.72
DeBERTa-V3-Large	Meme text	42.15	29.25	75.4
DeBERTa-V3-Large	VLM descriptions	42.93	30.18	74.36
XLM-R-Large	VLM descriptions	43.38	31.98	67.42
XGBoost	BLIP embeddings	47.55	79.79	33.87
Zero-shot VLM	Meme image & text	52.56	48.8	56.94
Early Fusion	All	67.84	88.93	54.83